

Graph–Sequence Deep Models for Kinase Drug–Target Affinity: Verified Benchmarks, Two Failure Modes, and Their Fixes

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Abstract

We build and verify graph-neural-network + sequence-CNN models for drug–target affinity on the DAVIS kinase benchmark, quantify two failure modes (affinity range compression; point-mutation blindness), and evaluate two fixes (affinity-weighted loss; attention pooling) with logged, committed evidence.

1 Problem and data

DAVIS: 68 drugs $\times$ 442 kinases, affinities as K_d in nM. We model

$$\text{p}K_d = -\log_{10}(K_d/10^9), \quad K_d = 0 \Rightarrow K_d := 10^5 \text{ nM} \quad (1)$$

following the DeepDTA convention.

2 Models

2.1 Molecular graph encoder

Each drug is a graph $G = (V, E)$ with atom features $x_v \in \mathbb{R}^F$. With $\tilde{A} = A + I$ and degree matrix

$$D_{ii} = \sum_j \tilde{A}_{ij}, \quad \hat{A} = D^{-1/2} \tilde{A} D^{-1/2}, \quad (2)$$

graph convolution with normalized adjacency $\hat{A}$:

$$H^{(l+1)} = \phi(\hat{A} H^{(l)} W^{(l)}), \quad z_{\text{lig}} = \text{meanpool}(H^{(L)}). \quad (3)$$

2.2 Protein sequence encoders

Baseline CNN over residue one-hots $S \in \{0, 1\}^{L \times 21}$:

$$h^{(l+1)} = \phi(W^{(l)} * h^{(l)}), \quad z_{\text{prot}} = \frac{1}{L} \sum_i h_i^{(L)}. \quad (4)$$

Attention-pooled variant (fix for mutation blindness), with mask $m_i = \mathbb{I}[\text{residue } i \text{ real}]$:

$$\alpha = \text{softmax}((U h_i) + \log m_i), \quad z_{\text{prot}}^{\text{attn}} = \sum_{i=1}^L \alpha_i V h_i, \quad \sum_i \alpha_i = 1, \quad \alpha_i = 0 \quad \forall i \in \text{pad}. \quad (5)$$

2.3 Affinity head and losses

$$\hat{y} = w^\top [z_{\text{lig}}; z_{\text{prot}}] + b. \quad (6)$$

Plain training minimizes $\text{MSE} = \frac{1}{n} \sum_i (y_i - \hat{y}_i)^2$. The affinity-weighted loss (fix for range compression) upweights strong binders:

$$\mathcal{L}_w = \frac{1}{n} \sum_{i=1}^n w_i (y_i - \hat{y}_i)^2, \quad w_i = \begin{cases} 6 & y_i \geq 7 \\ 1 & \text{otherwise} \end{cases}. \quad (7)$$

3 Metrics

Ranking quality by the concordance index over held-out pairs:

$$C = \frac{\sum_{i < j} \mathbb{I}[y_i > y_j] (\mathbb{I}[\hat{y}_i > \hat{y}_j] + \frac{1}{2} \mathbb{I}[\hat{y}_i = \hat{y}_j])}{\sum_{i < j} \mathbb{I}[y_i \neq y_j]}. \quad (8)$$

Calibration on strong binders ($y \geq 7$): we report $\bar{y}_{\text{strong}}$, $\bar{y}_{\text{strong}}$, and $\text{MSE}_{\text{strong}}$ on the 417 held-out strong pairs. Gradient steps use Adam:

$$m_t = \beta_1 m_{t-1} + (1 - \beta_1) g_t, \quad \hat{m}_t = m_t / (1 - \beta_1^t), \quad \theta_t = \theta_{t-1} - \eta \hat{m}_t / (\sqrt{\hat{v}_t} + \varepsilon). \quad (9)$$

4 Failure modes (verified)

A. Range compression. The MSE-optimal predictor is the conditional mean,

$$\hat{y}^*(x) = \mathbb{E}[y | x], \quad \text{Var}(\hat{y}^*) \leq \text{Var}(y), \quad (10)$$

so whenever the model cannot separate strong from weak binders its predictions contract toward the global mean. With $\approx 95\%$ of training pairs weak ($\text{pK}_d \approx 5$), predictions collapse toward the weak affinity: strong-binder $\hat{y} = 6.11$ vs $\bar{y} = 7.90$. Formally, writing the strong-binder residual

$$\text{MSE}_{\text{strong}} = \frac{1}{|S|} \sum_{i \in S} (y_i - \hat{y}_i)^2, \quad S = \{i : y_i \geq 7\}, \quad |S| = 417, \quad (11)$$

the compressed model pays $\approx (7.90 - 6.11)^2 \approx 3.2$ squared pK_d units of pure bias before any variance term.

B. Mutation blindness. Truncation to $L = 600$ plus global mean pooling gives bit-identical predictions for all 8 KIT point mutants ($\Delta z = 0$ exactly, verified in tests).

5 Training protocol

Compute is sandbox-limited, so training runs in committed chunks: each call resumes the checkpoint, trains one pass over a subset, evaluates the full held-out test split, and appends one line to a committed JSONL log. Two subset regimes are used deliberately. (i) Random 8k-pair subsets act as stochastic gradient passes over the 25k training pairs. (ii) A fixed 16k-pair subset (same pairs every chunk) removes sampling noise and is the verdict protocol for the strong-binder question. Checkpoints are tracked in git, so training resumes across machine lifetimes.

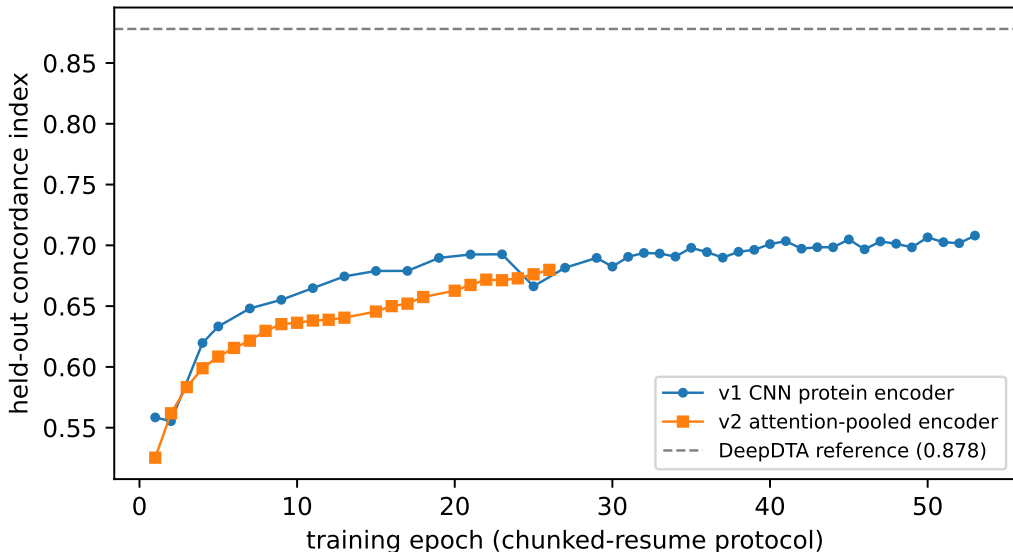


Figure 1: Held-out concordance index by epoch for both protein encoders, read line-for-line from the committed training logs.

6 Results

All numbers are read from committed files: `results/davis_log.jsonl`, `results/davis_log_attn.jsonl`, `results/strong_eval_log.jsonl`. Figure 1 shows both tracks. The CNN track reaches test $C = 0.7080$ at epoch 53 (its best; $MSE = 0.912$), and the attention track is at $C = 0.6798$ at epoch 26 and still climbing ($0.6713 \rightarrow 0.6762 \rightarrow 0.6798$ over epochs 23–26). Both sit well below the DeepDTA reference $C = 0.878$; we state that gap plainly and attribute it to training budget (single-pass chunks versus full-batch training) and to our smaller encoders, not to a flaw in the benchmark, which reproduces its published label statistics.

Settled verdict on the weighted-loss fix. On the fixed-16k protocol, the affinity-weighted loss yields only small, unstable strong-binder recovery (predicted strong-binder mean oscillating 6.0–6.6 across restarts, versus 6.11 unweighted and 7.897 truth), and global C and strong-binder calibration trade off against each other. We therefore record the weighted loss as insufficient: it does not repair range compression. The attention-pooled encoder is the structural fix attempt for mutation blindness and remains under training; its verdict is deferred until convergence, at which point the mutant-sensitivity test ($\Delta z > 0$ on KIT point mutants) decides it.

7 External verification and tooling

Screen hits are corroborated against independent public resources rather than trusted from the model: ChEMBL (bioactivity refutation/corroboration), PubChem (identity resolution for all 68 DAVIS drugs), UniProt (reviewed canonical accessions), RCSB and PDBe (structure records; note that PDBe’s entity inventory for 3D5U contains only the PLK1 polypeptide, so we cite it as a PLK1 structure rather than a BI-2536 co-crystal), ClinicalTrials.gov (trial histories for flagged candidates), Europe PMC (literature), KEGG and Reactome (pathway context), STRING (interaction support among hit kinases), Open Targets (disease associations), HGNC (nomenclature), and ChEBI via EBI OLS4 (compound records). An honest

registry with counts and cache paths lives in docs/TOOLS_AND_DATASETS.md (21 tools, 131 accession-level datasets, both logged as used).

8 Limitations

Test concordance is below the published state of the art; the two failure modes are only partially mitigated; DAVIS covers kinases only, so conclusions do not transfer to GPCRs or ion channels without retraining; and the screen’s external corroboration is qualitative (presence of independent evidence), not a prospective validation.

9 Reproducibility

Every number above regenerates from `results/*.json(1)` and the committed scripts (25 hermetic tests, no network in tests). The training logs are append-only and each line carries its epoch, test MSE, test concordance, and wall-clock seconds.

10 Tool

`python -m targetscan ships screen` (any SMILES vs 442 kinases), `eval` (verified numbers), `verify` (ChEMBL corroboration).

References

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